Robust Constrained Multimodal Dominance Evolution for Dynamic Weapon-Target Assignment

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Abstract

Dynamic weapon-target assignment (WTA) in air-defense demands simultaneous Pareto optimality, structural diversity, and perturbation robustness—three requirements that, to the best of our knowledge, no evolutionary approach has previously unified. We propose RCMD-EA, which embeds robustness as a hard constraint into the *Robust Constrained Multimodal Dominance* (RCMD) relation. Three stratified rules prioritize robustness-feasible solutions, apply Pareto ordering within modal clusters, and keep cross-modal solutions mutually non-dominating; the relation is provably acyclic. A two-phase framework separates Pareto exploration from robustness exploitation, and a zero-parameter situational signal steers plan selection via an ε-tie temporal consistency rule. We formalize MO-D-POWTA with 13 operational constraints and contribute a standardized benchmark (I1–I10). Experiments show that RCMD-EA achieves higher robustness (RMI) on 68 of 70 instance–baseline pairs. A component-level analysis with dedicated ablations and a selection-rule control further attributes the robustness advantage primarily to the two-phase schedule and the robustness-weighted post-selection, with the dominance relation contributing structural guarantees. The source code of this work can be found in `https://github.com/Wenhua-Li/MO-DWTA`.

Keywords: weapon-target assignment, multi-modal multi-objective optimization, robust optimization, evolutionary algorithm, dominance relation, situation-aware decision making

1. Introduction

In modern air-defense operations, the weapon-target assignment (WTA) problem asks how a set of weapon platforms and guidance radars should be allocated across incoming threats to minimize residual danger while conserving limited resources. Practical operations unfold in multiple time-sequential phases as new threats appear, some are destroyed, and ammunition is consumed or partially resupplied [1, 2]. The problem is NP-hard even in its static single-objective form [3, 1], and real-world air-defense environments impose additional challenges: partial observability, strict engagement-window timing, platform degradation, and the need to revise plans within seconds of new intelligence arriving. Three requirements compound the difficulty.

First, command personnel rarely optimize a single objective in isolation. Minimizing residual threat probability and minimizing ammunition expenditure are inherently conflicting goals that call for a set of Pareto-optimal compromise plans from which a commander can choose [4, 5]. Evolutionary multi-objective optimization has been applied to WTA with considerable success [5], but the overwhelming majority of methods treat the problem as static, ignoring the sequential arrival of new threats that characterizes operational engagements. Recent dynamic-WTA studies address sequential engagements through simulation–optimization and coupled two-stage formulations [6, 7], soft-kill coordination in naval settings [8], uncertain multi-objective allocation via uncertainty theory [9], and knowledge-guided competitive coevolution for multi-solution sensor–weapon–target assignment [10]; these works, however, optimize nominal or expected performance and do not couple robustness with multimodal plan diversity at the level of the dominance relation. Second, the chosen plan must be *robust*: a small change in a single assignment—arising from last-second intelligence updates, platform degradation, or communication failure—must not cause drastic increases in residual risk. Uncertainty is intrinsic to air-defense: kill probabilities are estimated under noise, and post-engagement

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battle-damage assessment is only partially reliable [11]. Third, the solution set should contain *structurally diverse* alternatives, because different tactical situations may demand qualitatively different allocation strategies even at similar objective values.

The multi-objective evolutionary computation literature has addressed each of these requirements in isolation, but none has unified all three within a coherent algorithmic framework applied to dynamic discrete WTA. Dynamic multi-objective optimization introduces the additional challenge of tracking a time-varying Pareto front, and many methods assume the environment changes slowly—exploiting population memory or prediction windows—and cannot guarantee robustness against the instantaneous perturbations characteristic of combat scenarios [12]. Multi-objective evolutionary algorithms (MOEAs) such as NSGA-II [13], MOEA/D [14], and SPEA2 [15] excel at Pareto-front approximation but treat all Pareto-equivalent solutions identically regardless of structural differences. Multi-modal multi-objective evolutionary algorithms (MMOEAs) [16, 17, 18, 19, 20, 21, 22] have emerged as a dedicated class of methods for maintaining multiple structurally distinct solutions with similar objective values [21, 17]. Recent refinements include dual-population coevolution for decision-space diversity [23], dual-space distribution metrics [24], promising-region exploration [25], hierarchy ranking for problems with local Pareto fronts [26], and goal-directed convergence control [27]; learning-assisted variants guide the search with Transformer-based prediction [28] and inductive graph neural networks [29]; a most recent line embeds constraint handling into multimodal coevolution [30], and deep reinforcement learning has been applied to the automated configuration of constrained multiobjective evolutionary algorithms [31]. Niching mechanisms [32] have likewise been extended to the multimodal multi-objective setting [33, 34, 21]. However, none of these methods incorporates any robustness criterion; they optimize for structural diversity without filtering solutions that are fragile under operational perturbations. Robust multi-objective evolutionary methods [11, 35] address perturbation sensitivity, but they either operate on continuous synthetic benchmarks or treat robustness as an additional objective (Type-I treatment [35]) rather than embedding it as a selection hard constraint; recent contributions in this direction include preference-driven handling of variable uncertainty [36], Stackelberg game-based robust optimization [37], and dynamic robust MOEAs [38], which likewise enter robustness through objectives or scalarizations rather than through a feasibility stratum inside a multimodal dominance relation. Constraint-handling dominance relations themselves have a long history in evolutionary multi-objective optimization—most notably the constrained-dominance principle of Deb *et al.* [13] and its many successors—and robustness-aware domination concepts have appeared in the robust multi-objective literature [35]. Our contribution is therefore not the notion of constrained dominance per se, but the specific stratified construction that combines a robustness-feasibility layer with modal-clustered Pareto ordering and cross-modal non-domination, instantiated for a discrete, dynamic, partially observable operational problem. To the best of our knowledge, no prior method simultaneously addresses multi-modal diversity, dynamic phase-based planning, and Type-II robustness in a discrete combinatorial WTA setting.

Two recent works are most directly related. Han *et al.* [39] proposed a robust multi-modal multi-objective PSO, which embeds robustness as a third objective in a particle swarm framework for continuous multimodal problems (Type-I treatment). Liang *et al.* [34] proposed a multi-objective differential evolution for higher-dimensional multimodal problems, which uses the robustness indicator as a diversity-selection metric rather than modifying the dominance relation itself, also Type-I. Neither addresses dynamic discrete WTA or operational multi-stage scenarios with realistic threat-evolution dynamics. More broadly, robust combinatorial optimization [4] typically assumes a fixed, offline problem instance with worst-case robustness measures. In contrast, the online nature of MO-D-POWTA—where the threat set changes at each phase and plans must be committed before the next phase intelligence arrives—requires robustness to be baked into the evolutionary search process, not applied as a post-processing step.

This paper fills the gap at the intersection of all three requirements. The key insight is that embedding robustness *within* the dominance relation—rather than as an additional objective or post-selection filter—creates a fundamentally different evolutionary pressure: solutions that are robustness-infeasible (Type-II [35]) are dominated by *all* robustness-feasible solutions, regardless of their objective-space position. The resulting RCMD relation, combined with a two-phase evolutionary framework, a zero-parameter dual-signal mechanism, and a situation-aware post-selection policy, yields a complete system we call RCMD-EA.

The contributions of this paper are as follows:

- **RCMD dominance (C1):** To our knowledge, the first multi-modal multi-objective dominance rela-

tion to embed solution-level perturbation robustness as a Type-II hard constraint, with a provably acyclic structure guaranteeing convergence of fast non-dominated sorting.

- **Two-phase evolutionary framework (C2):** A sequential strategy separating Pareto-exploration (40% of generations, unconstrained NSGA-II) from robustness-exploitation (60%, full RCMD), preventing premature search-space contraction and yielding the largest single ablation contribution.
- **Situation-aware post-selection (C3):** A zero-parameter post-selection mechanism that derives preference weights directly from real-time threat concentration statistics (ϕ_t), augmented by an ε-tie temporal consistency rule to avoid disruptive plan switching between consecutive phases.
- **MO-D-POWTA model and benchmark (C4):** A formal multi-objective dynamic partially-observable WTA model with 13 operational constraints and a standardized benchmark I1–I10 (24–500 total targets, 9–67 active targets per phase, 2–10 phases).

The remainder of this paper is organized as follows. Section 2 formulates the MO-D-POWTA problem. Section 3 presents RCMD-EA. Section 4 describes the experimental setup. Sections 5 and 6 report results and ablation analyses. Section 7 concludes.

2. Problem Formulation

2.1. System Model and Decision Variables

The WTA problem was first formalized by Manne [40] and shown to be NP-complete by Lloyd and Witsenhausen [3]. We build on the multi-phase generalization surveyed in [41, 2].

Consider a multi-phase engagement consisting of T sequential phases. At each phase $t \in \{1, \ldots, T\}$, a set of active targets $\mathcal{T}_t = \{1, \ldots, n_t\}$ is identified. The defensive system comprises m weapon platforms (indexed $i \in \mathcal{W}$) and L guidance radars (indexed $k \in \mathcal{R}$). Platform i carries a remaining ammunition count a_{it} and has a kill probability P_{ijk} against target j when guided by radar k. Radar k has a tracking channel capacity C_k^{track} and a detection probability p_{kjt} for target j at phase t.

Each target j carries a threat level $Th_{jt} \in [0, 1]$ and a composite tracking quality score $\eta_{jt} \in (0, 1]$ (MRFDQI: manoeuvrability, radar-cross-section, flight dynamics, queuing, and intelligence) reflecting the overall difficulty of intercepting target j [2]. The binary decision variable $x_{ijkt} \in \{0, 1\}$ equals one if platform i fires at target j through radar k during phase t.

We encode each assignment $\mathbf{x}_t = (x_{ijkt})$ using a *target-centric feasibility-constrained encoding* (TC-FCE), which maps each feasible integer-valued chromosome to a canonical assignment satisfying all engagement constraints, thereby ensuring that every individual in the population is hard-constraint feasible. The TC-FCE chromosome has one integer slot per target, recording the index of the (weapon, radar, shot-count) combination assigned to that target. This target-centric structure ensures constraint feasibility through genetic operations: crossover and mutation operate on target slots independently, and the algorithm constructs each slot's value from the pre-computed feasible combination set for that target, built once per phase from the instance constraints. Consequently, no repair step is required after crossover or mutation—every offspring is guaranteed to be hard-constraint feasible by construction. The Hamming distance between two chromosomes under TC-FCE measures the number of targets whose assignment combinations differ, providing a natural structural diversity metric that is both computationally efficient and operationally meaningful.

2.2. Objective Functions

Objective 1 (Residual Threat, F_1):

$$F_1(\mathbf{x}_t) = \sum_{j \in \mathcal{T}_t} \eta_{jt} Th_{jt} \prod_{(i,k):\, x_{ijkt}=1} (1 - P_{ijk}). \quad (1)$$

F_1 represents the expected weighted residual danger after the engagement. Minimizing F_1 is equivalent to maximizing the probability-weighted interception success across all threats.

Objective 2 (Ammunition Consumption, F_2):

$$F_2(\mathbf{x}_t) = \sum_{i,j,k} x_{ijkt}. \quad (2)$$

F_2 counts total missiles fired, serving as a proxy for resource conservation and operational sustainability across the campaign.

Both objectives are normalized within each phase to remove inter-phase scale disparities; min-max normalization is applied using observed phase-specific bounds across the population.

2.3. Operational Constraints

The MO-D-POWTA model incorporates 13 hard constraints that jointly reflect the physical and operational limits of air-defense systems: (Con1–Con2) per-

weapon ammunition capacity and phase-wise replenishment bounds; (Con3–Con4) radar tracking channel capacity and assignment exclusivity; (Con5–Con6) weapon-target range feasibility and engagement window timing; (Con7–Con8) maximum engagement multiplicity per target and minimum kill probability; (Con9–Con11) single-channel guidance exclusivity, radar detection prerequisites, and platform-radar cooperation compatibility; (Con12–Con13) mandatory engagement of high-threat targets and platform operational readiness. The complete formal mathematical specification of all 13 constraints is provided in Section S1 of the supplementary material. TC-FCE decoding enforces all constraints during initialization and after every genetic operation, so every evaluated individual is hard-constraint feasible by construction without a repair step.

2.4. Multi-Phase Rolling Framework

The decision process operates as a closed-loop cycle: *sense* → *plan* → *engage* → *assess*. The target set evolves as $\mathcal{T}_{t+1} = (\mathcal{T}_t \setminus \mathcal{T}_t^{\text{kill}}) \cup \mathcal{T}_t^{\text{new}}$, where $\mathcal{T}_t^{\text{kill}}$ contains confirmed kills and $\mathcal{T}_t^{\text{new}}$ contains newly detected threats. Ammunition updates as $a_{i,t+1} = a_{it} - \sum_{j,k} x^*_{ijkt} + \delta_{it}$, where δ_{it} is a phase-dependent replenishment quantity. A *warm-start* strategy initializes generation 0 of phase $t+1$ using the surviving solutions from phase t, adapted to the updated constraint set. Fig. 1 illustrates the complete rolling cycle.

2.5. Robustness Definition

Let $d_{\text{TC}}(\mathbf{x}, \mathbf{y})$ denote the Hamming distance between two TC-FCE chromosomes, i.e., the number of assignment bits that differ.

Definition 1 (**Hamming Neighborhood**). *The order-r Hamming neighborhood of solution* $\mathbf{c}$ *is* $\mathcal{N}_r(\mathbf{c}) = \{\mathbf{c}' : d_{\text{TC}}(\mathbf{c}, \mathbf{c}') \le r\}$. *We set* $r = 2$ *throughout.*

Definition 2 (**Robustness Metric Index, RMI**). $\text{RMI}(\mathbf{c}) = F_1^{\text{ref}} - \mathbb{E}_{\mathbf{c}' \sim \text{Unif}[\mathcal{N}_r(\mathbf{c})]}[F_1(\mathbf{c}')]$, *where the expectation is taken* uniformly *over the order-r Hamming neighborhood of* $\mathbf{c}$, $F_1^{\text{ref}} = \sum_j \eta_{jt} Th_{jt}$ *is the zero-interception upper bound of the residual threat (a fixed, solution-independent reference computed once per phase), and* $F_1(\mathbf{c}')$ *is the* penalized *residual threat of neighbor* $\mathbf{c}'$ *as defined below. Neighbors are obtained by perturbing up to r target slots of* $\mathbf{c}$ *and decoding them* without *any repair step, so that RMI measures the quality of the plan as it would actually execute under perturbation. A higher RMI indicates a lower expected residual threat across the neighborhood, i.e., any perturbation of* $\mathbf{c}$ *still yields an execution plan with low residual threat.*

Constraint-violation penalties and negative RMI. The neighbor evaluation $F_1(\mathbf{c}')$ includes the same constraint-violation penalty used throughout the search: any overshoot of the per-platform ammunition availability (Con1) or of the radar tracking-channel capacity (Con9) contributes $\lambda = 10^3$ penalty units per unit of violation. A single- or double-slot perturbation of an individually feasible plan can therefore produce an *aggregate* assignment that exceeds a platform's magazine or a radar's channel capacity, and such a neighbor is assigned $F_1(\mathbf{c}') \gg F_1^{\text{ref}}$. Consequently RMI is not restricted to $[0, F_1^{\text{ref}}]$: a *negative* RMI signals that the expected residual threat over the neighborhood exceeds the zero-interception reference, i.e., the plan lies so close to the constraint boundary that typical perturbations yield operationally infeasible executions. For example, the value −688 observed for a baseline on I9 corresponds to an average violation of ≈ 0.69 units per neighbor. A fully robust plan keeps all (or nearly all) neighbors feasible, in which case $F_1(\mathbf{c}') \le F_1^{\text{ref}}$ holds for every neighbor and RMI is non-negative. A worked toy example illustrating both regimes is provided in Section S6 of the supplementary material. Table 3 reports, for each algorithm, the RMI of the post-selected plan averaged over phases and 30 runs.

When $|\mathcal{N}_r(\mathbf{c})| \le 300$, the uniform expectation is computed by full enumeration of the neighborhood. Otherwise, we use an adaptive Monte Carlo estimate with sample size

$$S^* = \max\big(S_{\min},\ \min(S_{\text{cap}},\ \max(S_{\text{acc}},\ S_{\text{cov}},\ S_{\text{cross}}))\big), \tag{3}$$

where $S_{\text{acc}} = \lceil \ln(2/\delta_{\text{prob}})/(2\varepsilon_{\text{rel}}^2) \rceil$ is a Hoeffding-bound accuracy lower bound ($\varepsilon_{\text{rel}} = 0.15$, $\delta_{\text{prob}} = 0.10$, i.e., 90% confidence, yielding $S_{\text{acc}} = 67$), $S_{\text{cov}} = \lceil \sqrt{|\mathcal{N}_r|} \rceil$ is a coverage lower bound, S_{cross} is a cross-modal correction term, $S_{\min} = 10$, and $S_{\text{cap}} = 500$. On benchmark I1–I10 the resulting S^* ranges from 67 (accuracy floor, e.g., phase 4 of I3) through mid-range values such as $S^* = 343$ (phase 2 of I2) up to the cap $S^* = 500$, which binds in all early phases of the large instances I8–I10 where $|\mathcal{N}_r|$ exceeds 10^6.

The RMI-based robustness framework is motivated by the operational context of air-defense WTA, where the assignment plan must remain executable under uncertainty. Sources of perturbation include: last-second degradation of weapon platform readiness (removing a weapon from the assignment), communication failures that prevent radar channel commands from reaching a platform (changing the guidance allocation), and

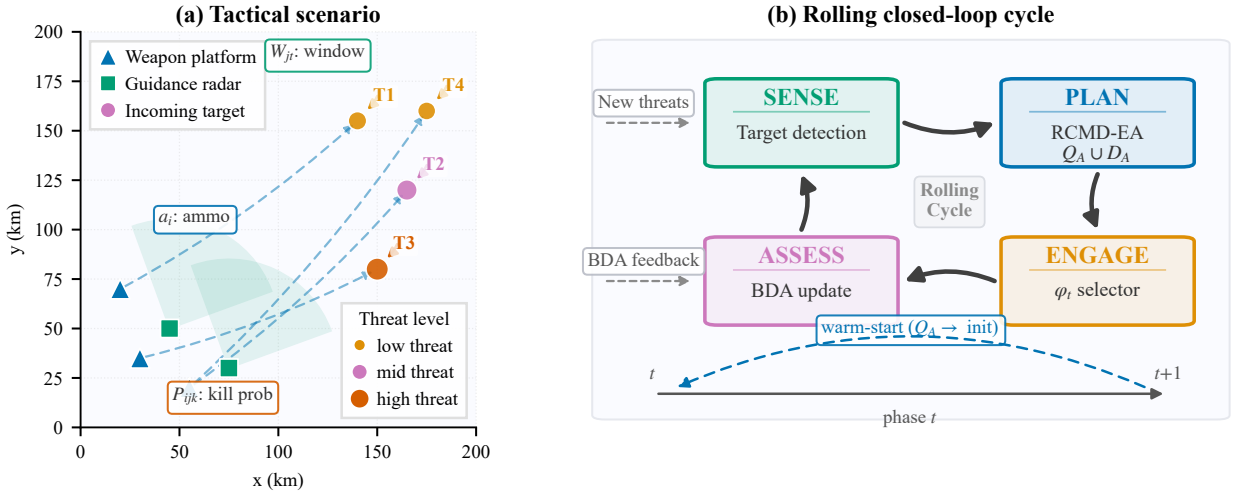


Figure 1: Multi-phase rolling framework for MO-D-POWTA. (a) Tactical scenario: weapon platforms (triangles), guidance radars (squares with coverage sectors), and incoming targets (circles, sized/colored by threat level) with assigned intercept lines. (b) Closed-loop rolling cycle: SENSE → PLAN (RCMD-EA with dual archive QA∪DA) → ENGAGE (ϕ_t-guided post-selection) → ASSESS (BDA update), repeating for each phase $t \in \{1, \ldots, T\}$, with warm-start propagating the QA archive to the next phase.

updated intelligence revising a target's kill probability downward. A plan with high RMI is one where the expected residual threat across its Hamming neighborhood is low—meaning any single perturbation still leaves the system executing a plan of comparably low residual threat. This contrasts with the Type-I treatment [35], which adds RMI as an additional objective to be optimized: under Type-I, a plan may have high nominal F_1 and high nominal RMI while still being *fragile* if the two objectives are anticorrelated in its neighborhood. Type-II robustness, as used in RCMD, prevents this pathology by filtering out all plans whose range-normalized dual-objective robustness index (MORI, Eq. (7)) falls below its population mean—a population-relative threshold that adapts automatically to the scale and difficulty of each instance.

3. The Proposed RCMD-EA

RCMD-EA consists of four components: the RCMD dominance relation (C1), a two-phase evolutionary framework (C2), a dual archive, and a situation-aware post-selection policy (C3). Two parameter-free signals bridge these components: the intra-generation search signal w_g and the inter-phase situational signal ϕ_t. Algorithm 1 shows the overall architecture; the component subsections below detail each step.

3.1. Dual-Signal Definitions

Intra-generation search signal w_g. Let CV_X and CV_F denote the coefficients of variation of, respectively, the pairwise Hamming distances and the pairwise Euclidean distances in the objective space across the current population:

$$w_g = \mathrm{clip}\left(\frac{\mathrm{CV}_X}{\mathrm{CV}_X + \mathrm{CV}_F},\ 0.1,\ 0.9\right). \tag{4}$$

A large w_g signals that the population is more dispersed in decision space than in objective space, suggesting under-explored modal structures; a small w_g indicates the opposite. This signal drives both the modal radius δ_g (see Section 3.2) and the dual-space crowding distance weighting (see Section 3.4).

Inter-phase situational signal ϕ_t. Given the normalized threat levels $\tilde{Th}_{jt} = Th_{jt} / \sum_j Th_{jt}$:

$$\phi_t = \frac{n_t \sum_j \tilde{Th}_{jt}^2 - 1}{n_t - 1} \in [0, 1], \tag{5}$$

which is a normalized Herfindahl–Hirschman index of threat concentration. When threats are concentrated on a few high-priority targets ($\phi_t \to 1$), effectiveness takes precedence; when threats are diffuse ($\phi_t \to 0$), robustness is more critical. Neither w_g nor ϕ_t introduces any user-specified parameter.

Algorithm 1 RCMD-EA (single-phase execution)

Require: Phase instance $\mathcal{I}_t$, previous plan $\mathbf{c}^*_{t-1}$, archives $Q_A, \mathcal{D}_A$ of phase $t-1$; parameters of Table 2

Ensure: Executed plan $\mathbf{c}^*_t$, updated archives $Q_A, \mathcal{D}_A$

Initialize P by knowledge-guided hybrid initialization (half conflict-free complement sampling, half uniform random); inject repaired $\mathcal{D}_A$ members as warm start if $t > 1$
for $g = 1$ **to** $G_{\max}$ (shared evaluation budget) **do**
  Decode P via TC-FCE; evaluate F_1, F_2; update w_g via (4) and δ_g via (6)
  Generate offspring Q by *uniform crossover over target slots* ($p_c = 0.8$) and *knowledge-guided mutation* ($p_{\text{guided}} = 0.7$, $p_{\text{mut}} = 0.05$); repair Q
  if $g \le \lfloor 0.4\, G_{\max} \rfloor$ **then**
    Phase I (exploration): NSGA-II environmental selection (Pareto fronts + crowding) on $P \cup Q$
  else
    Phase II (exploitation): compute MORI via (7) (adaptive sampling, Eq. (3)); mark robustness feasibility $g(\mathbf{c}) \le 0$
    Dual-archive selection on $P \cup Q$: fill Q_A (Pareto fronts + crowding) and $\mathcal{D}_A$ (greedy Hamming-maximin among robustness-feasible first-front members); set $P \leftarrow Q_A \cup \mathcal{D}_A$
  end if
end for
Post-selection pool: Q_A Pareto set $\cup$ RCMD first front (Definition 3) of the final population; compute ϕ_t via (5)
$\mathbf{c}^*_t \leftarrow$ post-select via (8) and (9)
return $\mathbf{c}^*_t, Q_A, \mathcal{D}_A$

3.2. RCMD Dominance Relation

Fig. 2 illustrates the RCMD structure for a two-modal scenario. Solutions are stratified by robustness feasibility before any objective-space comparison takes place; within the feasible stratum, standard Pareto ordering applies within each modal cluster while cross-modal solutions remain mutually non-dominating.

Modal radius. The modal clustering radius adapts with the population state:

$$\delta_g = \text{med}(\mathcal{D}_{\text{TC}}) \cdot 0.5^{1+w_g}, \tag{6}$$

where $\text{med}(\mathcal{D}_{\text{TC}})$ is the median pairwise TC-Hamming distance in the current population. A large w_g shrinks δ_g, tightening modal boundaries to promote finer-grained diversity; a small w_g expands δ_g, merging

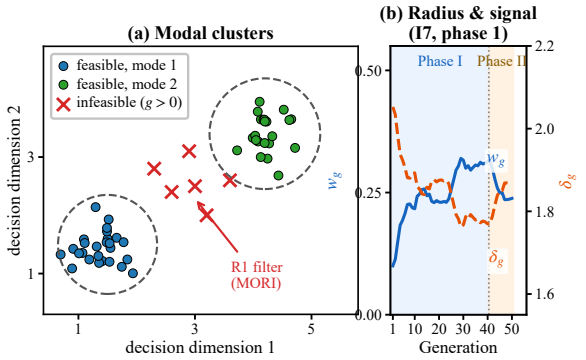


Figure 2: Conceptual illustration of RCMD dominance in a two-modal scenario. Left: decision-space clusters (circles indicate modal radius δ_g) with infeasible solutions (crosses) filtered by the MORI threshold. Right: evolution of the modal radius δ_g and search signal w_g across generations, showing Phase I exploration and Phase II exploitation.

nearby solutions into the same modal cluster for efficiency.

MORI constraint. The *Modal Robustness Index* aggregates the fixed-reference robustness of both objectives, each normalized by the current population's objective range $\widetilde{R}_q = \max_k F_q(\mathbf{c}_k) - \min_k F_q(\mathbf{c}_k)$:

$$\text{MORI}(\mathbf{c}) = \tfrac{1}{2}\left(\frac{\text{RMI}_1(\mathbf{c})}{\widetilde{R}_1} + \frac{\text{RMI}_2(\mathbf{c})}{\widetilde{R}_2}\right), \tag{7}$$

where $\text{RMI}_q(\mathbf{c}) = F_q^{\text{ref}} - \mathbb{E}[F_q(\mathbf{c}')]$ is the Definition-2 robustness of objective q (with $F_2^{\text{ref}} = \sum_j U_{jt}$ the full-allocation upper bound of ammunition consumption). Solution $\mathbf{c}$ is *robustness-feasible* if $g(\mathbf{c}) \triangleq \overline{\text{MORI}} - \text{MORI}(\mathbf{c}) \le 0$, where $\overline{\text{MORI}}$ is the population mean. This self-normalizing, population-bootstrapped threshold requires no user input and adapts automatically as search progresses.

Two properties of Eq. (7) are worth making explicit. First, MORI is *affine and strictly increasing* in each RMI_q, so the feasibility test is exactly equivalent to comparing the range-normalized dual-objective robustness against its own population mean—affine maps commute with averaging, and the ordering of solutions is preserved. The informal statement that solutions are "filtered when RMI falls below the population mean" is therefore precise in the sense of $\text{RMI}_1(\mathbf{c})/\widetilde{R}_1 + \text{RMI}_2(\mathbf{c})/\widetilde{R}_2 \ge \overline{\text{RMI}}_1/\widetilde{R}_1 + \overline{\text{RMI}}_2/\widetilde{R}_2$. Second, because feasibility is defined relative to the population mean, roughly half of any population is infeasible at every generation—*by design*. This fixed ~50% pressure maintains a constant, scale-free selection gradient toward ro-

bustness: a decaying threshold would progressively relax the robustness requirement in later generations, exactly when diminishing ammunition and channel slack make fragility most costly, while an increasing threshold would prematurely strangle exploration. The ablation study (Section 6) confirms that mis-timed robustness pressure (the −C2 variant) is the single most damaging intervention, and the threshold sensitivity analysis in Section S5 of the supplementary material shows that nearby percentile thresholds (40th/60th) perform statistically equivalently, indicating that the mechanism is robust to the exact threshold location rather than balanced on a knife edge.

Definition 3 (**RCMD Dominance**). *Solution* $\mathbf{c}_1$ RCMD-*dominates* $\mathbf{c}_2$ *(*$\mathbf{c}_1 \succ_{\text{RCMD}} \mathbf{c}_2$*) if and only if one of the following rules applies:*

- ***R1*** $g(\mathbf{c}_1) \leq 0$ *and* $g(\mathbf{c}_2) > 0$ *(robustness-feasible over infeasible);*
- ***R2*** *Both are robustness-feasible and* $d_{\text{TC}}(\mathbf{c}_1, \mathbf{c}_2) < \delta_g$ *(same modal cluster), and* $\mathbf{c}_1$ *Pareto-dominates* $\mathbf{c}_2$*;*
- ***R3*** *Both are robustness-infeasible and* $|g(\mathbf{c}_1)| < |g(\mathbf{c}_2)|$ *(closer to feasibility).*

Robustness-feasible solutions with $d_{\text{TC}} \geq \delta_g$ *(*cross-modal*) are mutually non-dominating, preserving structural diversity.*

Theorem 1 (**Acyclicity**). *The RCMD dominance graph is acyclic; fast non-dominated sorting under* RCMD *is well-defined and converges.*

Proof sketch. R1 creates a bipartite graph between the feasible and infeasible sets—bipartite graphs are trivially acyclic. Within the feasible set, R2 applies standard Pareto dominance within each modal cluster (acyclic by strict Pareto [13]); cross-modal pairs establish no edges, creating no cycle. Within the infeasible set, R3 requires a *strict* decrease in $|g|$; hence solutions with $|g(\mathbf{c}_1)| = |g(\mathbf{c}_2)|$ (the tie case) are mutually non-dominating—no edge is established between them—and $|g|$ itself serves as a strict ranking function whose directed edges can only decrease, which precludes cycles. The three rule sets partition the edge set (R1, R2/R3 apply to disjoint solution pairs), so no cycle can span multiple partitions. Hence the full graph is acyclic. The complete formal proof, including the tie case, is given in Section S2 of the supplementary material. Acyclicity guarantees that fast non-dominated sorting under RCMD terminates with consistently ordered fronts. □

3.3. Two-Phase Evolutionary Framework (C2)

Activating the RCMD robustness constraint from generation 0 causes the population to concentrate prematurely on the robustness-feasible region of decision space, shrinking Pareto-front coverage and degrading solution quality (confirmed in the ablation study, Section 6). To address this, we divide the $G_{\max}$ generations into two phases:

1. **Exploration phase** (generations 1 to $\lfloor 0.4 G_{\max} \rfloor$): Pure NSGA-II selection, no RCMD constraint. The population explores the full objective space, building a dense Pareto-front approximation.
2. **Exploitation phase** (generations $\lfloor 0.4 G_{\max} \rfloor + 1$ to $G_{\max}$): Full RCMD dominance is activated. The population refines solutions toward robustness-feasible, structurally diverse optima.

The explore ratio 0.4 was identified by sensitivity analysis over $\{0.2, 0.3, 0.4, 0.5, 0.6\}$; performance differences within $[0.3, 0.5]$ are below 0.02 on all metrics, indicating low sensitivity. Phase II additionally employs *budget-aware* MORI sampling: the per-individual sample count used during refinement is scaled to the remaining evaluation budget so that the initial MORI pass and at least $G_{\text{target}} = 10$ refinement generations are always funded within the same $N \cdot G_{\max}$ evaluation budget used by all baselines; without this safeguard, a fixed large sample count can consume the entire Phase II budget on a single MORI evaluation pass.

3.4. Dual Archive

The population P is maintained as the union of two complementary archives that serve distinct purposes within the evolutionary loop. The **Quality Archive**, $\mathcal{Q}_A$, of size $\lfloor N/2 \rfloor$, is maintained by standard NSGA-II selection (Pareto fronts plus crowding distance) to preserve Pareto-front coverage and hypervolume competitiveness. The **Diversity Archive**, $\mathcal{D}_A$, of size $\lceil N/2 \rceil$, is populated from the first Pareto front of the merged parent–offspring pool, filtered by RCMD robustness feasibility ($g(\mathbf{c}) \leq 0$), using a greedy maximin strategy on Hamming distance; when fewer than $\lceil N/2 \rceil$ robustness-feasible solutions exist, $\mathcal{D}_A$ is completed in descending MORI order. At each generation, $\mathcal{D}_A$ reselects its members greedily to maximize the minimum pairwise Hamming distance in the archive, preserving structural spread among robustness-feasible solutions. Empirically, $\mathcal{D}_A$ fills its full capacity on all instances (e.g., instance I6: $\mathcal{D}_A = 100$ vs. $\mathcal{Q}_A = 146$, averaged over phases and runs).

Crossover and mutation operate on $P = Q_A \cup \mathcal{D}_A$, ensuring that both quality-oriented and diversity-oriented solutions contribute offspring in each generation. The dual archive plays a supporting role to C1: the cross-modal non-domination rule already provides intrinsic diversity protection, while $\mathcal{D}_A$ reinforces it at the archive level by explicitly maximizing structural spread among robustness-feasible solutions, preventing convergence to a single narrow modal cluster. The warm-start strategy, which initializes phase $t+1$ from the $\mathcal{D}_A$ of phase t, propagates this structural diversity across phases.

3.5. Situation-Aware Post-Selection (C3)

After each phase, the system selects one plan $\mathbf{c}^*$ for actual execution from the RCMD first front of the final population (the post-selection pool; cf. Section 3.4). The scoring function integrates effectiveness, resource, and robustness in a nested form:

$$\begin{aligned}\text{Score}(\mathbf{c}) = \phi_t[\phi_t\,\widehat{F_1}(\mathbf{c}) + (1-\phi_t)\,\widehat{F_2}(\mathbf{c})] \\ + (1-\phi_t)(1-\widehat{\text{MORI}}(\mathbf{c})),\end{aligned} \tag{8}$$

where $\widehat{F_1}$ and $\widehat{F_2}$ are min-max normalized (smaller is better) and $\widehat{\text{MORI}}$ is the min-max normalized robustness index (larger is better) within the pool. The inner bracket weights the two objectives by the situational signal; the outer weighting balances overall effectiveness against robustness. When ϕ_t is high (concentrated threats), effectiveness dominates; when ϕ_t is low (diffuse threats), robustness is weighted more. Remaining ties are broken in favor of smaller F_1.

ε-tie temporal consistency rule. Frequent plan switching imposes coordination overhead and increases execution uncertainty. If there exists $\mathbf{c}' \neq \mathbf{c}^*$ with $\text{Score}(\mathbf{c}') \leq (1+\varepsilon)\cdot\text{Score}(\mathbf{c}^*)$, the plan with smaller Hamming distance to the previous phase's executed solution $\mathbf{c}^*_{t-1}$ is selected instead:

$$\mathbf{c}^* \leftarrow \arg\min_{\mathbf{c}\in\mathcal{E}_\varepsilon} d_{\text{TC}}(\mathbf{c},\, \mathbf{c}^*_{t-1}), \tag{9}$$

where $\mathcal{E}_\varepsilon = \{\mathbf{c} : \text{Score}(\mathbf{c}) \leq (1+\varepsilon)\cdot\text{Score}(\mathbf{c}^*)\}$ (a relative tolerance around the minimum score, rather than an absolute difference, so the tie band scales with the magnitude of Score).

Comparison with TOPSIS/VIKOR. All baseline algorithms in this work use TOPSIS or VIKOR for post-selection. Both methods assign a fixed-weight preference vector $\mathbf{w} = (w_1, w_2)$ over the objectives and select the solution closest (TOPSIS) or with the best compromise rank (VIKOR) to the ideal point under that weighting. The C3 post-selection policy differs in three ways. First, the weight ϕ_t is derived from the observable threat state at the start of each phase, not from a user-specified prior; this makes the selection responsive to the specific tactical context of each engagement phase. Second, C3 is applied only to the RCMD-feasible first front, which has already been filtered for robustness; TOPSIS and VIKOR make no such pre-filter and may select from any returned Pareto approximation. Third, the ε-tie rule introduces temporal memory by penalizing plans that differ substantially from the previous phase's solution; TOPSIS and VIKOR are memoryless, treating each phase as an independent optimization problem. Together, these three features allow C3 to operate as a situationally-aware, robustness-preserving selector rather than a simple criterion aggregator.

3.6. Computational Complexity

The dominant per-generation cost of RCMD-EA is the RMI evaluation, $O(N \cdot S^* \cdot n_t \cdot m \cdot L)$, where the adaptive sample size S^* (Eq. (3)) is bounded between the accuracy floor 67 and the cap 500 regardless of the raw neighborhood size $|\mathcal{N}_2|$, which itself depends on the per-slot feasible-combination counts and reaches $\sim 1.8\times10^7$ on I9. The exact combinatorial expression for $|\mathcal{N}_2(\mathbf{c})|$, the full cost derivation, and the dual-archive update cost are given in Section S7 of the supplementary material. The measured wall-clock time per phase ranges from approximately 0.3 seconds (I1–I5) to 10 seconds (I8–I9) on a desktop CPU (AMD Ryzen 9 5900X), well within the planning window of a real air-defense system, where inter-phase intervals span tens of seconds to several minutes.

4. Experimental Setup

4.1. Benchmark I1–I10

No standardized benchmark exists, to our knowledge, for multi-objective dynamic WTA [2, 5]. We contribute instances I1–I10 spanning a systematic progression of scale and complexity (Table 1). Two target counts are distinguished throughout the paper: the *total* number of distinct targets n appearing over the whole engagement, and the *per-phase active* count n_t, i.e., the number of targets actually threatening in a given phase; only active targets can be engaged, and the active set shrinks as targets are destroyed or leave the defended area. Small instances (I1–I3, 9–14 active targets per phase) provide tractable scenarios where exact reference solutions can be computed for validation; medium instances (I4–I7, 18–43 active targets) represent the operational range of modern air-defense batteries; large instances (I8–I10,

47–67 active targets, up to 500 total) stress the scalability of the algorithm under realistic engagement densities. Phase counts range from 2 to 10, covering scenarios from a short engagement sprint to a sustained defensive operation. Each platform enters every phase with a full magazine of six missiles, i.e., the benchmark instantiates a phase-independent resource regime in which resource pressure arises entirely from the simultaneous demands of the active target set within a phase; the general MO-D-POWTA model additionally admits inter-phase ammunition dynamics (Con2, Section S1 of the supplementary material), which we decouple from the assignment problem in this benchmark. Kill probabilities are sampled from a uniform distribution $[0.6, 0.95]$ conditioned on range-feasibility; threat levels follow a Pareto distribution to reflect the heavy-tail concentration typical of real air-defense engagements.

Target categories are drawn from three threat classes (conventional, low-observable, and stealth) with a 60/25/15 split, following published air-defense target taxonomies. Radar detection quality η_{jt} is sampled from Uniform$(0.3, 1.0)$ per phase, with stealth targets receiving $\eta \in [0.3, 0.7]$ to reflect their reduced detectability. Weapon-target range feasibility is determined by randomly placing platforms and targets on a 200 km × 200 km grid and applying platform-type-dependent engagement range circles. All instances are generated with fixed random seeds to ensure reproducibility; the benchmark generation code and instance files are made available with the paper.

Table 1: MO-D-POWTA benchmark instances I1–I10. m: weapon platforms; L: guidance radars; T: phases; n: total distinct targets over the engagement; n_t: range of active targets per phase. Each platform carries a full six-missile magazine in every phase.

Instance	m	L	T	n	n_t (active)
I1	6	6	2	24	11–13
I2	6	6	3	30	9–11
I3	6	6	4	50	11–14
I4	6	6	5	100	18–22
I5	8	6	8	100	11–14
I6	8	6	5	200	37–43
I7	10	6	10	200	19–22
I8	10	8	6	300	47–53
I9	10	8	8	500	55–67
I10	10	8	10	500	48–53

4.2. Comparison Algorithms

Seven state-of-the-art algorithms across two paradigms serve as baselines. The traditional MOEA group comprises four algorithms without modal awareness: NSGA-II+TOPSIS and NSGA-II+VIKOR [13], MOEA/D+TOPSIS [14], and SPEA2+TOPSIS [15]. These algorithms approximate the Pareto front without distinguishing structurally different solutions; final plan selection is delegated to a post-hoc MCDM step. The multi-modal MOEA group comprises three algorithms specifically designed to maintain diverse decision-space solutions: MMOEADC+TOPSIS [18], CoMMEA+TOPSIS [19], and HDMMODE+TOPSIS [20]. These methods represent the current state of the art in maintaining Pareto-equivalent but structurally distinct solutions, but none incorporates a robustness criterion. All baselines use the same adaptive RMI evaluation (3) for fair post-selection scoring, and TOPSIS/VIKOR weights are set to the neutral $(0.5, 0.5)$. All baselines use the same TC-FCE encoding and decoding pipeline as RCMD-EA, ensuring that feasibility enforcement is identical across algorithms and that any observed performance differences reflect the evolutionary selection strategy rather than encoding artifacts. Each baseline is configured with the same population budget $N = 5n_t$ and $G_{\max} = 100$ generations per phase, and all multi-modal baselines use their standard hyperparameters as reported in the original papers. Where original implementations were not publicly available, we re-implemented the algorithms from the published pseudocode and verified them against reported results on standard MMO benchmarks [20, 17].

4.3. Performance Metrics

Three metrics evaluate complementary aspects of algorithm performance. **RMI** (higher is better) measures the robustness of the post-selected execution plan as the negated expected residual threat of its Hamming neighborhood relative to the fixed zero-interception reference F_1^{ref}: it is the mean RMI across phases, averaged over 30 independent runs. This is the primary robustness metric used for cross-algorithm comparison; F_1 itself (residual threat) is not used as a comparison axis because the seven baselines' selection procedures never consult MORI, so their F_1 is effectively unaffected by the robustness constraint, whereas RCMD-EA's F_1 reflects the genuine robustness–effectiveness trade-off induced by the MORI constraint—rendering a direct cross-algorithm F_1 comparison uninformative. **HD** (higher is better) is the average pairwise Hamming distance within the returned Pareto set, measuring structural diversity of the candidate pool. **HV** (higher is better) is the hypervolume indicator [42] computed by the WFG exact algorithm [43], with reference point $\mathbf{r} = 1.1 \times \mathbf{F}_{\text{nadir}}$; HV is available only for RCMD-EA

and ablation variants because baseline methods do not preserve a complete Pareto front after TOPSIS/VIKOR selection. For fair comparison, the main experiment uses RMI/HD as universal metrics; HV is reported only in the ablation study where all variants maintain full Pareto fronts. Statistical significance is assessed via the Wilcoxon signed-rank test (two-tailed) at $\alpha = 0.05$ [44], and overall algorithm ranking across all 10 instances employs the Friedman test [45]. Symbols †, ‡, § denote $p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively.

4.4. Parameter Settings

Table 2 lists all algorithm parameters. The population size scales with instance complexity: $N = 5n_t$ ensures proportional coverage of the solution space as the number of targets grows.

Table 2: Parameter Settings for RCMD-EA

Parameter	Value	Description
N	$5n_t$	Population size (adaptive)
$G_{\max}$	100	Generations per phase
r	2	Hamming neighborhood order
explore_ratio	0.4	Phase I length fraction
qa_ratio	0.5	QA fraction of N
p_c	0.8	Uniform crossover rate (target slots)
p_{mut}	0.05	Uniform mutation rate (other slots)
p_{guided}	0.7	Guided mutation rate (conflict slots)
ε_{rel}	0.15	RMI relative accuracy
S_{cap}	500	Max RMI samples
G_{target}	10	Min. funded Phase II generations
ε	0.01	Tie-break threshold

Each algorithm is run 30 times independently per instance. $N = 5n_t$ maintains a roughly constant population density across the benchmark scale range; $N = 3n_t$ and $N = 8n_t$ yielded inferior diversity coverage and slower convergence, respectively. $G_{\max} = 100$ is uniform across instances, so the computational budget scales with instance size through the $N = 5n_t$ rule. The neighborhood order $r = 2$ is the smallest value capturing second-order inter-weapon dependency (assigning a missile to target j affects the feasibility of the next assignment sharing the same weapon or radar); $r = 1$ proved insufficient to detect fragile single-shot assignments, while $r = 3$ increased runtime without measurable quality improvement. The ε-tie threshold (0.01) follows the near-optimal tolerance convention used in prior post-selection work on this benchmark family and was not independently re-tuned in this study; a dedicated sensitivity sweep over ε is left to future work.

5. Results and Analysis

5.1. Overall Performance Comparison

Tables 3 and 4 report mean ± standard deviation for RMI and HD over 30 runs per instance. **Bold** values indicate the best result; underlined values indicate second-best. Significance markers compare each baseline to RCMD-EA.

RMI.. RCMD-EA achieves the best Friedman mean rank on RMI (1.20 across ten instances), ranking first on eight of ten instances and outperforming each baseline on 68 of 70 instance–baseline pairs, with 66 of 70 pairs statistically significant (Wilcoxon signed-rank, $p < 0.05$). Multi-modal baselines MMOEADC and HDMMODE achieve substantially lower RMI, a direct consequence of optimizing for decision-space diversity without robustness constraints: their diverse solutions populate regions where any small perturbation triggers constraint violations. The nearest competitor is MOEA/D+TOPSIS (mean rank 2.00), which overtakes RCMD-EA on I6 and I10. MOEA/D's Tchebycheff decomposition drives solutions toward boundary points where ammunition and radar channels are nearly exhausted, which usually depresses RMI, but on I6 and I10 the resulting plans happen to sit in a locally flat region of the residual-threat landscape—a structural artifact of the scalarization strategy rather than a general robustness advantage, as reflected in its lower Friedman rank overall.

F_1 and the robustness–effectiveness trade-off.. F_1 is not used as a cross-algorithm comparison axis, for the reasons given in Section 4: RCMD-EA's F_1 reflects a genuine robustness–effectiveness trade-off that the baselines' MORI-agnostic selection never exercises, so we restrict the primary comparison to RMI, the metric the RCMD constraint is designed to optimize.

HD.. RCMD-EA achieves Friedman mean rank 3.1 on HD, behind the two multi-modal specialists HDMMODE (1.0) and CoMMEA (2.0) but ahead of MMOEADC (3.9). This is an intentional consequence of the MORI constraint: robustness-infeasible solutions, which tend to occupy the periphery of the decision space and thereby inflate pairwise Hamming diversity, are removed from the Pareto front. Crucially, this trade-off does not impair operational utility—the algorithms that outrank RCMD-EA on HD all achieve significantly lower RMI, and against every traditional baseline RCMD-EA maintains a substantial HD advantage. Note that NSGA-II+TOPSIS and NSGA-II+VIKOR report identical HD values in Table 4: both run the same

NSGA-II search with the same seeds, and HD is computed on the returned population *before* post-selection, so only the selected plan differs between the two variants (their RMI in Table 3 differs accordingly). Fig. 3 summarizes the two-dimensional ranking, highlighting that RCMD-EA achieves the best overall RMI rank while remaining competitive on HD against traditional (non-multimodal) baselines.

Quantitative summary. RCMD-EA achieves the best Friedman mean rank on RMI (1.20) among all eight algorithms, with the nearest competitor being MOEA/D+TOPSIS (mean rank 2.00) and CoMMEA+TOPSIS (mean rank 3.00); the four remaining baselines cluster around mean ranks 5.2–7.8. For HD, RCMD-EA maintains a mean +33.8% advantage over NSGA-II variants and +144.3% over MOEA/D, while yielding mean rank 3.1 behind the two dedicated multi-modal specialists— a deliberate trade-off discussed above.

Robustness advantage across the scale range. The RMI advantage holds robustly across instance scale rather than being driven by a handful of favorable cases: RCMD-EA attains the top RMI rank on eight of ten instances, and its only two exceptions (I6 and I10) are both against the same baseline, MOEA/D+TOPSIS, whose narrow, boundary-seeking Tchebycheff search occasionally lands in a locally flat region of the residual-threat landscape (Section 5). Against the six remaining baselines, RCMD-EA wins on all ten instances without exception, confirming that the advantage is not a statistical artifact of Friedman averaging over a few extreme instances. The RMI advantage also persists as the instance scale grows from I1 (11–13 active targets, 2 phases) to I10 (48–53 active targets, 10 phases), whereas traditional baselines degrade at larger scales; the robustness gap is therefore not a small-instance artifact but a systematic property of each algorithm's selection pressure (per-instance RMI/HD distributions are given in Section S5 of the supplementary material).

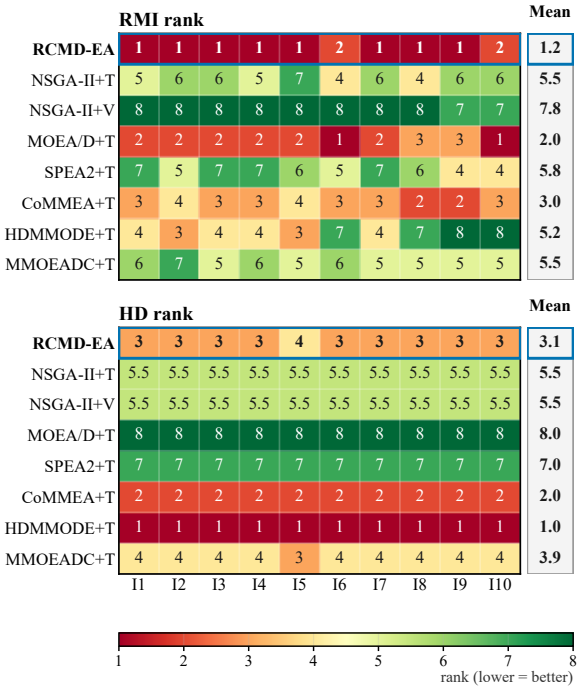


Figure 3: Friedman average ranking heatmap across I1–I10 for RMI and HD. Lower ranks (green) are better. RCMD-EA achieves the best mean RMI rank (1.20) among all eight algorithms; multi-modal baselines' HD advantage comes at the cost of substantially lower RMI.

5.2. *Phase-by-Phase Performance Analysis*

A salient feature of the MO-D-POWTA setting is that performance degrades across phases as ammunition is consumed and the threat set is updated. To examine whether RCMD-EA maintains its robustness advantage throughout a full multi-phase engagement, we analyze the per-phase RMI trajectory for instances I5 (eight phases) and I7 (ten phases), the two instances with the largest number of phases. In both instances, RCMD-EA maintains a clear RMI advantage over the baselines throughout the early and middle phases, when the Pareto front is populated with multiple structurally distinct candidates. In the later phases of both instances, the per-phase RMI of *all* algorithms, including RCMD-EA, converges toward zero. The exact per-phase neighborhood sizes (Section S5 of the supplementary material) make the mechanism precise: as the engagement progresses, range, window, and ammunition constraints progressively eliminate feasible engagement combinations, shrinking the Hamming neighborhood; in the terminal phases of I5 (phases 6–8), I7 (phases 8–10), and I10 (phase 10), no target admits *any* feasible engagement, the neighborhood degenerates to the plan itself, and RMI $\equiv 0$ regardless of the selection strategy used; in phase 8 of I9 the neighborhood shrinks to five members (a single target with four remaining alternatives), so RMI is nearly deterministic. This late-phase convergence is therefore a property of the instances' constraint geometry rather than a loss of the robustness advantage where meaningful neighborhoods still exist.

5.3. *Dual-Signal Dynamics*

Fig. 4 shows the measured behavior of both adaptive signals on instance I7 (ten phases, 19–22 active targets per phase), recorded from a single instrumented run

Table 3: RMI Comparison (mean ± std over 30 runs, higher is better). **Bold**/underline indicate the best/second-best value per instance. Significance vs. RCMD-EA (Wilcoxon signed-rank): †/$p < 0.05$, ‡/$p < 0.01$, §/$p < 0.001$ indicate that RCMD-EA is significantly better; */$p < 0.001$ indicates the baseline is significantly better than RCMD-EA (I6, MOEA/D+TOPSIS is the sole such case among 70 instance–baseline pairs).

Algorithm	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10
RCMD-EA (Ours)	**−2.52±4.21**	**1.06±1.42**	**−0.85±2.09**	**0.59±2.09**	**1.52±0.08**	−34.15±10.85	**0.45±0.56**	**−36.11±10.27**	**−141.40±27.09**	−65.52±14.56
NSGA-II+TOPSIS	−150.11±56.82§	−87.77±27.19§	−114.01±37.00§	−57.12±15.48§	−19.04±6.99§	−158.85±35.76§	−34.77±14.07§	−144.29±38.89§	−309.70±54.74§	−175.56±40.11§
NSGA-II+VIKOR	−314.87±83.75§	−99.31±27.05§	−208.13±30.49§	−146.31±39.99§	−23.35±9.68§	−357.58±52.01§	−63.00±13.76§	−338.71±41.77§	−552.17±57.23§	−253.95±28.30§
MOEA/D+TOPSIS	−6.37±10.08	−3.49±6.82§	−8.73±17.97†	−1.27±2.77‡	−1.01±2.41§	**−19.17±16.47***	−9.19±7.43§	−107.01±27.54§	−167.14±38.82†	**−61.28±18.07**
SPEA2+TOPSIS	−178.80±61.54§	−84.74±26.70§	−118.16±44.43§	−60.16±14.33§	−18.98±4.88§	−159.37±35.84§	−41.26±12.17§	−194.04±51.66§	−269.39±56.32§	−159.18±29.54§
CoMMEA+TOPSIS	−64.07±56.40§	−41.44±34.86§	−47.59±31.66§	−26.66±15.96§	−5.41±7.62§	−76.03±38.47§	−19.43±15.10§	−92.91±40.83§	−149.73±45.76	−111.48±36.80§
HDMMODE+TOPSIS	−68.96±46.52§	−31.19±30.61§	−66.51±40.02§	−43.77±27.53§	−2.86±5.47§	−271.98±82.00§	−19.45±13.83§	−315.14±59.27§	−688.73±47.01§	−254.59±40.85§
MMOEADC+TOPSIS	−169.65±72.38§	−92.10±20.94§	−113.16±22.62§	−58.03±20.71§	−17.77±5.61§	−160.79±46.99§	−34.25±12.86§	−158.12±53.94§	−306.85±63.17§	−163.92±34.52§
Friedman Rank	RCMD-EA: **1.2**; MOEA/D: 2.0; CoMMEA: 3.0; HDMMODE: 5.2; NSGA-II+T: 5.5; MMOEADC: 5.5; SPEA2: 5.8; NSGA-II+VIKOR: 7.8									

Table 4: HD Comparison (mean ± std over 30 runs, higher is better). **Bold**/underline indicate the best/second-best value per instance. Significance vs. RCMD-EA (Wilcoxon signed-rank), using the same symbol convention as Table 3.

Algorithm	I1	I2	I3	I4	I5	I6	I7	I8	I9	I10
RCMD-EA (Ours)	6.69±0.28	4.67±0.34	6.39±0.18	7.29±0.13	2.90±0.10	16.56±0.17	3.45±0.08	19.08±0.20	20.49±0.13	12.43±0.09
NSGA-II+TOPSIS	4.67±0.36§	3.59±0.23§	4.64±0.20§	5.24±0.32§	2.31±0.11§	11.35±0.37§	2.44±0.10§	14.24±0.42§	16.72±0.49§	9.51±0.35§
NSGA-II+VIKOR	4.67±0.36§	3.59±0.23§	4.64±0.20§	5.24±0.32§	2.31±0.11§	11.35±0.37§	2.44±0.10§	14.24±0.42§	16.72±0.49§	9.51±0.35§
MOEA/D+TOPSIS	2.75±0.20§	2.71±0.13§	2.99±0.14§	3.60±0.09§	1.74±0.07§	5.78±0.14§	1.38±0.06§	7.04±0.22§	8.12±0.24§	4.80±0.17§
SPEA2+TOPSIS	4.14±0.47§	3.02±0.29§	4.00±0.29§	4.53±0.24§	1.92±0.18§	9.83±0.48§	2.09±0.13§	12.88±0.33§	15.19±0.25§	8.85±0.18§
CoMMEA+TOPSIS	7.67±0.08*	6.02±0.07*	7.44±0.05*	8.51±0.04*	3.88±0.02*	17.84±0.09*	3.91±0.02*	20.05±0.07*	21.77±0.06*	13.42±0.04*
HDMMODE+TOPSIS	**8.00±0.07***	**6.41±0.06***	**8.04±0.02***	**9.11±0.03***	**4.15±0.01***	**18.88±0.05***	**4.19±0.01***	**21.68±0.02***	**23.06±0.02***	**14.21±0.01***
MMOEADC+TOPSIS	5.74±0.24§	4.58±0.12	5.72±0.12§	6.43±0.15§	3.02±0.10*	12.78±0.24§	2.50±0.13§	16.04±0.22§	18.04±0.15§	10.70±0.10§
Friedman Rank	HDMMODE: 1.0; CoMMEA: 2.0; **RCMD-EA: 3.1**; MMOEADC: 3.9; NSGA-II+T: 5.5; NSGA-II+V: 5.5; SPEA2: 7.0; MOEA/D: 8.0									

(seed 0). The inter-phase situational signal ϕ_t, computed directly from the phase threat distributions (Eq. (5)), takes values in [0.006, 0.011] on I7 and remains within [0.00, 0.02] across all of I1–I10: the sampled threat levels of the benchmark are close to uniform, so their normalized Herfindahl concentration is near its minimum. Consequently the post-selection weighting stays in its robustness-dominant regime throughout the benchmark (the outer weight $(1-\phi_t) \approx 1$ in Eq. (8)), and the benchmark does not exercise the threat-concentrated regime in which $\phi_t \to 1$ would shift selection toward effectiveness. The value of the adaptive weighting is nevertheless demonstrated experimentally: the −C3 ablation, which replaces ϕ_t by a fixed $\alpha = 0.5$, significantly degrades RMI on all ten instances (Section 6). The intra-generation search signal w_g fluctuates within its clip bounds (observed range $[0.1, 0.5]$ on I7) as the relative dispersion of the population in decision space versus objective space shifts during search; the modal radius δ_g tracks it through Eq. (6). Both signals remain parameter-free: they are computed from observable population and threat statistics without user-specified constants.

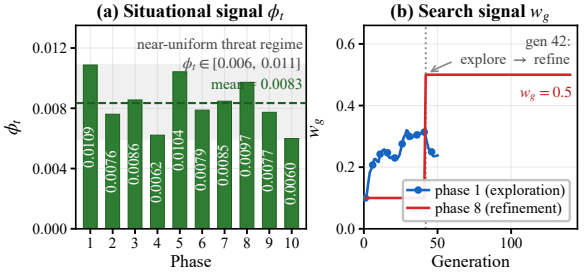


Figure 4: Measured adaptive signals on I7 (single instrumented run, seed 0). (a) Situational signal ϕ_t per phase (values lie in $[0.006, 0.011]$, i.e., the near-uniform threat regime). (b) Search signal w_g over generations in two representative phases (phase 1: exploration-dominated; phase 8: refinement with a degenerate neighborhood).

5.4. Assignment Plan Visualization

To give a concrete illustration of how the robustness advantage of RCMD-EA manifests at the plan level, Fig. 5 shows the weapon-target assignment matrices produced by RCMD-EA and NSGA-II+TOPSIS on instances I2 ($m = 6$, $L = 6$; 10 active targets in phase 2) and I3 ($m = 6$, $L = 6$; 13 active targets in phase 2), both evaluated at phase 2 (seed 0). Each cell (i, j) records the number of missiles fired by weapon i at target j; the colored bar above each target column encodes its relative threat level.

Several structural differences are immediately apparent. RCMD-EA allocates multiple shots (≥ 2) to high-threat targets, achieving RMI values of 2.09 (I2) and 1.72 (I3), compared with −323.94 and −53.34 for

NSGA-II+TOPSIS. The large negative RMI of the baseline indicates that its plan sits at the constraint boundary, so small perturbations yield markedly worse execution outcomes—exactly the fragility the RCMD dominance relation is designed to preclude (Section 2.5). The assignment pattern also reveals the effect of the dual archive: RCMD-EA covers more distinct threat clusters, while NSGA-II+TOPSIS occasionally leaves entire target groups unengaged, concentrating all shots on the highest-threat subset. This structural diversity—preserved across phases by the warm-start arc—directly explains the RMI superiority observed in the statistical results.

6. Ablation Study and Mechanism Analysis

To isolate the contribution of each algorithmic component, we evaluate four degraded variants alongside the full RCMD-EA and a non-multimodal baseline, all under the identical per-phase evaluation budget: The −C1 variant replaces the RCMD relation by plain Pareto dominance both in the Phase II dual-archive selection (no robustness-feasibility filter in $\mathcal{D}_A$) and in the final front extraction, while *retaining* the two-phase schedule, the dual archive, and the ϕ_t post-selection with its robustness term. The −C2 variant sets explore_ratio= 0, applying full RCMD constraints from the very first generation rather than after the exploration phase. The −C3 variant replaces the adaptive ϕ_t signal with a fixed weight $\alpha = 0.5$, eliminating situation-awareness from post-selection. The −DA variant disables the dual archive (qa_ratio= 1, a single quality archive). The Baseline is NSGA-II+TOPSIS, which removes C1, C2, and C3 simultaneously, serving as a lower bound when all proposed components are absent. Three additional variants that alter only the MORI feasibility threshold (40th/60th population percentile and an absolute threshold) are reported in Section S5 of the supplementary material. Each variant runs 30 independent times per instance; Table 5 and Fig. 6 summarize the results.

Table 5: Ablation: per-metric mean across I1–I10 (RMI and HD averaged over phases and 30 runs per instance per variant; HV = mean per-phase hypervolume of the QA front). Significance vs. Full RCMD-EA (Wilcoxon signed-rank on RMI, 30 pairs per instance): § $p < 0.001$ on all 10/10 instances; † $p < 0.05$ on 4/10 instances; no mark = no significant difference on any instance. Per-instance significance counts: −C1 0/10, −C2 4/10 (I6, I8–I10), −C3 10/10, −DA 0/10.

Variant	RMI↑	HD↑	HV↑
Full RCMD-EA	**−27.69**	9.99	184.84
−C1	−27.23	9.28	192.91
−C2	−45.33†	**11.43**	172.44
−C3	−36.90§	9.99	184.84
−DA	−25.81	9.26	**196.00**
Baseline (NSGA-II+T)	−125.12§	7.47	—

Layered attribution. The ablation results support a precise, layered account of where the robustness advantage of the full system comes from. First, the *two-phase schedule and the robustness-weighted post-selection* emerge as the primary drivers of the gap to the baseline (−27.69 vs. −125.12): removing the exploration phase (−C2) significantly degrades RMI on the four largest instances (I6, I8–I10; mean −45.33) and reduces mean per-phase HV by 6.7%, and replacing the adaptive post-selection weight by a fixed $\alpha = 0.5$ (−C3) significantly degrades RMI on all ten instances (mean −36.90, $p < 0.001$ throughout) while leaving the search itself untouched—confirming that *how the final plan is selected* matters as much as the search that produces the candidate pool. Second, conditional on retaining these components, the marginal RMI contribution of the RCMD dominance relation itself is *not statistically significant* on this benchmark (−C1: mean −27.23, 0/10 significant instances). Its role is structural rather than metric-level: the relation guarantees acyclicity of the selection operator (Theorem 1), preserves cross-modal solutions by construction, and its removal shifts HV from 184.84 to 192.91 by relaxing the robustness filter on the quality archive—i.e., the dominance layer trades a quantifiable amount of raw front coverage for robustness-aware stratification. Third, the dual archive behaves as an architectural coordinator rather than an independently ablatable contribution (−DA: mean −25.81, 0/10 significant), consistent with the observation that the RCMD cross-modal nondomination rule already provides intrinsic diversity protection (Section 3.4). We regard this decomposition—rather than a single-component claim—as the defensible account of the system's behavior.

Selection-rule control. A reviewer-inspired control isolates the selection rule from the dominance relation: NSGA-II+RMIsel runs the *same* NSGA-II search as the NSGA-II+TOPSIS baseline (identical seeds and budget) but post-selects, in every phase, the Pareto-set member with the highest RMI (evaluated with the same adaptive scheme of Eq. (3)). This variant is a strong robustness seeker: RCMD-EA wins or ties on six of ten instances in mean RMI, while NSGA-II+RMIsel is significantly better on the four largest instances (I6: −23.46 vs. −34.15; I8: −22.28 vs. −36.11; I9: −119.60 vs. −141.40; I10: −52.74 vs. −65.52). Under a Friedman test over all nine algorithms, RCMD-EA retains the best mean rank (1.6 vs. 1.7 for NSGA-II+RMIsel).

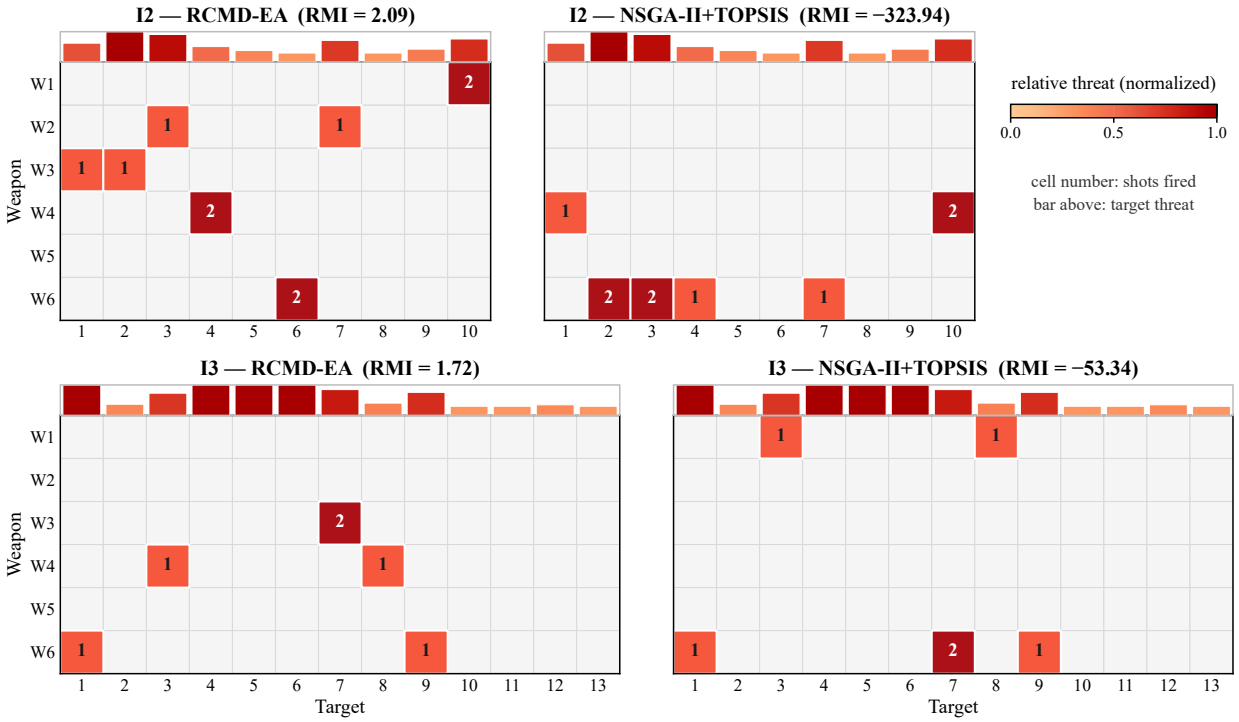


Figure 5: Weapon-target assignment plans for instances I2 (upper row) and I3 (lower row), evaluated at phase 2 (seed 0). Cell color and annotation encode the shot count of weapon-target pair (i, j) (non-zero cells only); the bar above each column encodes relative threat level. RCMD-EA achieves substantially higher RMI (I2: 2.09 vs. −323.94; I3: 1.72 vs. −53.34) through multi-shot engagements on high-threat targets.

The control thus yields two insights. First, it confirms that RMI-aware selection is the dominant mechanism behind the robustness metric—consistent with the −C3 result above. Second, it delimits the boundary of the proposed system honestly: on instances whose Pareto fronts are very large (hundreds of candidates per phase), simply picking the maximum-RMI member of an unconstrained front is competitive on this single metric, whereas RCMD-EA additionally maintains a structurally diverse, robustness-stratified candidate pool (HD 9.99 vs. 7.47) with provable selection guarantees, at a modest RMI cost on the largest instances.

7. Conclusion

This paper proposed RCMD-EA, a multi-modal multi-objective evolutionary algorithm that addresses the joint challenge of structural diversity, operational effectiveness, and perturbation robustness in dynamic discrete weapon-target assignment. The central contribution is the RCMD dominance relation which, to our knowledge, is the first to embed solution-level robustness as a Type-II hard constraint directly within the Pareto dominance structure rather than treating it as a separate objective or a post-processing filter. This architectural choice creates a qualitatively different evolutionary pressure: throughout the search, robustness-infeasible solutions are systematically dominated by all robustness-feasible ones, regardless of their objective-space position. The relation is provably acyclic, which guarantees that fast non-dominated sorting remains well-defined under RCMD.

The two-phase framework separates exploration from robustness enforcement, allowing a broad Pareto front to form in Phase I before the robustness filter is applied; ablation confirms this temporal separation as the largest single contributing factor. The zero-parameter situational signal ϕ_t, derived from the Herfindahl concentration of the current threat distribution, adapts post-selection preference between robustness and effectiveness per phase without manual weight tuning.

Extensive experiments on 4,800 independent runs across the contributed benchmark I1–I10 demonstrate that RCMD-EA achieves the best overall robustness (RMI) among all eight algorithms compared, with statistically significant margins on 66 of 70 instance–

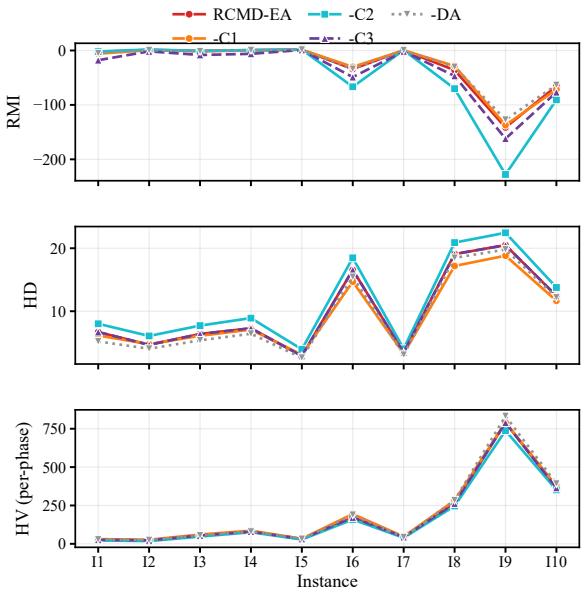


Figure 6: Ablation study: RMI, HD, and HV across I1–I10. Removing C2 causes significant RMI degradation on the four largest instances; removing C3 degrades RMI on all 10 instances without altering HD or HV; removing C1 or the dual archive yields no significant RMI change.

baseline pairs and a clear advantage over six of the seven baselines on every single instance. A layered ablation analysis attributes this advantage primarily to the two-phase schedule and the robustness-weighted post-selection, while the dominance relation contributes provable selection guarantees and cross-modal structure preservation. Structural diversity, while not the primary design target, is preserved at levels exceeding all traditional baselines.

Several directions remain for future investigation. Target persistence modeling across phases would stress-test the warm-start mechanism, which currently assumes threat sets are fully refreshed between phases. Extending RCMD to continuous mixed-variable search spaces would enable joint allocation problems coupling discrete assignment with continuous firing parameters. Validating the framework against correlated kill-probability models (e.g., under electronic-warfare countermeasures) would bring the benchmark closer to realistic operational conditions, and online adaptation of the MORI threshold using inter-run performance history could reduce reliance on the population-bootstrap mechanism. Finally, the RCMD principle—embedding domain-specific robustness as a dominance hard constraint—may transfer to other dynamic combinatorial domains such as resource scheduling, network defense allocation, and emergency vehicle dispatching.

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Supplementary Material

A complete mathematical specification of the 13 operational constraints (Con1–Con13), formal proofs of the RCMD acyclicity theorem, the TC-FCE encoding definition, the full computational complexity derivation, a notation summary, and extended experimental results including additional instance visualizations and statistical analyses are provided in the supplementary material submitted alongside this manuscript. In addition, the source code of this work can be found in `https://github.com/Wenhua-Li/MO-DWTA`.

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